

PPL

1 what is "this" keyword in java

- >refers to the current object of the class
- differentiate between global and local variables that have same name
- invoking the current class constructor
- invoking the current class method

4 what is an associative array.

- > is a data structure that stores key-value pairs
- also known as dictionary, hash table or map
- associative arrays unique keys of various types to access their corresponding values

6 what is subroutine calling sequence

- >a subroutine calling sequence refers to the specific set of steps
- refers to the order in which subroutines are called and returned from
- passing arguments to the subroutine
- saving the current state
- allocating memory
- transferring control
- restoring the previous state

7 what is difference between discrete and scalar types

- >discrete types - are types have a finite set of distinct values
- values are distinct and

countable

scalar types - are types that have a single value

single component data values

8 what is dynamic memory location

-> is a technique where memory is allocated and deallocated at runtime

is the process of allocating memory space during runtime

to manage data structure

9 explain different time at which binding decision are made

-> process of linking a variable or identifier to its definition or value

compile time -static binding

decisions made by the compiler before the programs runs

link time -decisions made by the linker when combining different

compiled libraries into a single executable program

load time -decision made by the os loader when the program is

loaded into main memory for execution

run time - dynamic binding

late binding

decisions postponed until the

program is actually running

10 explain different types of operator available in C

-> arithmetic - perform basic mathematical operation

+

addition, -subtraction, *multiplication,
/division, % modulus division

relational - used for comparison and return Boolean value(T/F)

== equal to, != not equal
to, >greater than,

<less than, >= greater than
or equal to

<= less than or equal to
logical - combine two or more
conditions

&& logical AND-true if both
operands are true

|| logical OR -true if at
least one operand is T

! logical NOT - reverse the
logical state

bitwise - perform operations at the
bit level

& bitwise AND, | bitwise OR, ~
bitwise NOT

<< left shift, >>right shift
assignment - assign values to variables

==

increment decrement - increase or decrease
a variables value by 1

++ increment, --decrement, ++a
prefix, a++ postfix

conditional - if else

11 what is dangling reference ?how to avoid
dangling reference problem

-> dangling reference occurs when a pointer
points

to memory that has been deallocated or
reused

access or modify memory through dangling
pointer

deallocation of memory

returning address of local variables

variable going to out of scope

avoid-

set pointers to NULL after freeing memory

avoid returning addresses of local

variables

limit pointer scope

use memory debugging tools

12 give the syntax and use of following
function

I) getchar

-> reads a character from standard input

syntax - char var=getchar()

I) putchar

-> writes a character to standard output
syntax - putchar(char variable)

13 why are there so many programming languages

-> bcoz different languages are suited for different tasks

each language has its own strengths and weaknesses

languages support different ways of thinking about problems

such as object-oriented, functional, or procedural language

15 explain heap based allocation

-> heap based allocation is a memory management technique where

memory is allocated and deallocated from a heap data structure

memory is allocated from the heap during the program execution

heap allocation dynamically program execution

work :

dynamic allocation

runtime management

manual or automatic deallocation

16 explain any two parameter passing modes with suitable example

-> call by value - a copy of the actual arguments value is passed

to the function parameter
changes made to the
parameter inside the function
do not affect the
original variable in the calling code
this method prevents
modification of the original data

eg

```
#include<stdio.h>
void increment(int x)
{
    x=x+1;
    printf("inside the
function x = %d ",x);
}
int main()
{
    int num=10;
    printf("before
function call num=%d",num);
    increment(num);
    printf("after
function call num=%d",num);
}
```

call by reference - a reference to the
original argument is passed to the function
it avoids the creating
copy

changes made to the

parameter inside the function

do affect the original
variable in the calling code

```
eg #include<stdio.h>
void increment(int *x)
{
    *x=*x+1;
    printf("inside the
function *x = %d ",*x);
}
int main()
{
    int num=10;
    printf("before
function call num=%d",num);
    increment(num);
    printf("after
function call num=%d",num);
}
```

call by name

17 explain two techniques of garbage
collection

-> is a memory management technique that
automatically frees memory

occupied by objects that are no longer
referenced

reference counting -

this technique adds a counter to each

heap allocated object

to track the no. of pointers referring to it

the counter is incremented when a new pointer references the object

decremented when a pointer stops referencing it

mark and sweep -

this is a batch oriented technique that runs when available free space

it operates in two phases the mark phase identifies all objects and

the sweep phase iterates through heap

this approach can cause to pause during collection but can handle

cyclic data structure correctly

19 what is nesting of classes in java

explain with suitable example 6-5

nesting of classes in java refers to defining a class within another class

they can access only static member of the outer class

inner class have access all members

static nested class are like regular top level class

nesting class divided into 2 categories :

non static nested classes

static nested classes

eg

```

public class outerClass
{
    public static class nested
    {
        public void disp()
        {
            sop("msg");
        }
    }
    public class innerClass
    {
        public void disp()
        {
            sop("msg");
        }
    }
}

pvsm(string a[])
{
    outerClass.nested obj = new
outerClass.nested();
    obj.disp();

    outerClass outerobj = new
outeClass();
    outerClass.innerClass innerobj =
outerobj.new innerClass();
    innerobj.disp();
}

```

NOTES

19. Dataflow language provide inherently parallel model justify true or false

TRUE

in a dataflow lang an instruction executed as its required

input data become available sequentially
program counter

multiple instructions executed simultaneously

20. List any 2 function of c pre-processor
file inclusion - it handles directives like #include

replace directive with the special content, source file

macro expansion - it replaces macro names defined with #define with

their corresponding values before the main compilation

21. What is binding time

binding time is a point in the program execution life cycle when a

name, data, entity associated with their property

binding time is a range from lang design time to compile time,
load time, runtime.

24. What is memorization

to speed up computer programs

to manage storage

it is optimization technique used to speed up comp programs

25. What is heap dynamic array

it is an array whose storage is allocated on the heap at runtime.

its size not fixed at compile time.

it can changed and modified during program execution.

in lang like c++ this is managed using pointers and dynamic

memory allocation functions

28. Define polymorphism with example6-13

many forms

is a concept in object oriented programming

it enables a single interface to represent many forms or types

eg

```
class animal
```

```
{
```

```
    void speak()
```

```
    {
```

```
        sop("animals make a sound")
```

```
    }
```

```
}
```

```
class dog extends animal
```

```
{
```

```
    void speak()
```

```

        {
            sop("dog barks")
        }
    }
    class cat extends animal
    {
        void speak()
        {
            sop("cat meow")
        }
    }

```

```

animal mydog = new dog()

```

```

mydog.speak()

```

```

animal mycat = new cat()

```

```

mycat.speak()

```

a base class animal have method speak().

derived classes dog,cat can override

this method to produce different sound

demonstrate polymorphic behavior

32. What is static chain

a static chain is used in block scoped programming lang to access global variables

it is chain of pointer stored in the subprogram

33. What is union 4-25

a union is a special data structure

allows different data types stored in same memory location

the size of union is depend by size of

its largest no.

it is a special data type in available lang like c,c++.

34. Describe the three string length options 4-9

static length- the length is fixed at compile time

cannot change during execution

eg C(char a[20])

limited dynamic length - the string length vary during runtime

up to a predefined maximum limit

memory location is static but the used portion is dynamic

eg pascal(versions),A

dynamic length-both the maximum size and current length change arbitrarily

during the execution

eg std::string